# 1 Mathematical Background

This chapter collects the mathematical material needed later for randomized least-squares methods and for covariance-based dependence correction. The main ingredients come from four directions: classical least-squares theory, random projections and subspace embeddings, symmetric positive definite operators and their matrix functions, and perturbation questions related to conditioning. Although all spaces in this thesis are finite-dimensional, it is often convenient to use the language of linear maps, ranges, and subspaces rather than only matrix coordinates.

For an overdetermined system $A \in \mathbb{R}^{n\times d}$ with $n \geq d$, the least-squares problem

$$\min_{x\in\mathbb{R}^d} \|Ax - b\|_2$$

asks for the vector whose image under $A$ is closest to $b$ in Euclidean norm. Geometrically, this means projecting $b$ onto the range of $A$. If $A$ has full column rank, the minimizer is unique. In exact arithmetic, it may be characterized by the normal equations or, more stably, by an orthogonal factorization.

The randomized viewpoint replaces the original problem by a smaller one. A sketching matrix $S \in \mathbb{R}^{s\times n}$, with $s$ much smaller than $n$, is chosen so that the geometry of the range of $A$ is approximately preserved after compression. One then solves

$$\min_{x\in\mathbb{R}^d} \|SAx - Sb\|_2.$$

The quality of this replacement depends on how well the sketch preserves norms and angles on the relevant subspace. This is where subspace embeddings and the Johnson–Lindenstrauss phenomenon enter the picture. The sketch should act almost isometrically on the subspace that carries the least-squares geometry. The formal definitions and embedding guarantees are given in ??→p.??.

Two sketch families are especially important in this thesis. The first is the Gaussian sketch, whose entries are independent random variables with suitable scaling. The second is the class of structured sketches, especially transforms based on randomized Hadamard matrices. Gaussian sketches are the cleanest theoretical baseline. Structured sketches are more attractive computationally because they can exploit fast transforms.

Conditioning plays a central role throughout. Even when sketching gives a good approximation in principle, the numerical behavior of the reduced problem depends strongly on the conditioning of $SA$, that is, on the stability of the compressed operator on the relevant subspace. This becomes even more important in finite precision, since rounding errors may be harmless for a well-conditioned problem and severe for an ill-conditioned one. The relation between sketching quality and conditioning is discussed in ??→p.??.

The later mixed-precision discussion therefore rests on a simple principle. Randomized approximation error and floating-point error should not be studied in isolation. A useful theory must account for both at the same time.

## 1.1 Singular Value Decomposition and Pseudoinverse

The basic structural tool for least-squares analysis is the thin singular value decomposition. If $A \in \mathbb{R}^{n\times d}$ has rank $r$, then

$$A = U_r \Sigma_r V_r^\top,$$

where $U_r \in \mathbb{R}^{n\times r}$ and $V_r \in \mathbb{R}^{d\times r}$ have orthonormal columns and

$$\Sigma_r = \operatorname{diag}(\sigma_1, \dots, \sigma_r), \qquad \sigma_1 \geq \cdots \geq \sigma_r > 0,$$

contains the nonzero singular values. This factorization separates the geometry of the range, the nullspace, and the spectral scales of the operator.

If $A$ has full column rank, the least-squares minimizer may be written as

$$x_\star = A^\dagger b, \qquad A^\dagger = (A^\top A)^{-1} A^\top = V_r \Sigma_r^{-1} U_r^\top,$$

where $A^\dagger$ is the Moore–Penrose pseudoinverse. This formula is useful for analysis because it makes explicit that the small singular values of $A$ are the source of sensitivity: directions associated with $\sigma_j \ll 1$ are amplified by $\sigma_j^{-1}$.

The operator norm of $A$ is

$$\|A\|_2 = \sigma_1,$$

and for a full-column-rank matrix the Euclidean condition number is

$$\kappa_2(A) = \frac{\sigma_1}{\sigma_r}.$$

This quantity measures how strongly data perturbations can be amplified by the least-squares solve and reappears later in the analysis of sketched systems.

## 1.2 Normal Equations, QR, and SVD

The least-squares problem admits several algebraic formulations, but they do not behave equally well numerically. The normal equations

$$A^\top A x = A^\top b$$

are attractive because they reduce the problem to a square system. Their drawback is that they square the spectrum:

$$\sigma_j(A^\top A) = \sigma_j(A)^2, \qquad \kappa_2(A^\top A) = \kappa_2(A)^2.$$

Thus a moderately ill-conditioned least-squares problem can become a severely ill-conditioned linear system when written in this form.

By contrast, if $A = QR$ is a thin QR factorization with $Q$ having orthonormal columns and $R \in \mathbb{R}^{d\times d}$ upper triangular, then the least-squares problem reduces to

$$\min_x \|Rx - Q^\top b\|_2,$$

or equivalently to the triangular solve

$$Rx = Q^\top b$$

in the full-column-rank case. QR avoids the explicit formation of $A^\top A$ and is therefore the standard stable dense formulation.

The SVD is more expensive than QR but more informative. It reveals rank deficiency directly, yields the pseudoinverse explicitly, and makes clear how solution sensitivity depends on the singular spectrum. In the present thesis, QR is the practical dense baseline, while the SVD is the conceptual reference that makes the conditioning questions transparent [1].

## 1.3 Least-Squares Perturbation Viewpoint

The numerical sensitivity of least squares depends on more than the matrix norm alone. Let $x_\star$ solve

$$\min_x \|Ax - b\|_2,$$

and write $r_\star = b - Ax_\star$ for the optimal residual. Classical perturbation theory shows that under small perturbations $\Delta A$ and $\Delta b$, the relative change in the minimizer is controlled by the condition number together with a term that reflects the size of the residual. Schematically,

$$\frac{\|\Delta x_\star\|_2}{\|x_\star\|_2} \lesssim \kappa_2(A)\left(\frac{\|\Delta A\|_2}{\|A\|_2} + \frac{\|\Delta b\|_2}{\|b\|_2}\right) + \kappa_2(A)^2 \frac{\|r_\star\|_2}{\|A\|_2\,\|x_\star\|_2}\,\frac{\|\Delta A\|_2}{\|A\|_2}\,,$$

with more precise constants depending on the chosen formulation Ch. 1 [1]. The important qualitative message is that solution sensitivity is mild when the residual is small and $A$ is well conditioned, but can be much worse in inconsistent or ill-conditioned problems.

This observation explains why residual bounds and solution-error bounds should be kept conceptually distinct. A method may preserve the least-squares objective accurately while still producing a noticeably perturbed minimizer if the problem is geometrically sensitive. That distinction becomes central later when sketching and floating-point perturbations are analyzed together.

## 1.4 Floating-Point Recap for Least Squares

The mixed-precision chapter develops the full floating-point model later, but the least-squares background already needs one simple principle. Forming the reduced operator, the normal matrix, or the orthogonal factorization in finite precision does not solve the exact problem posed by $A$ and $b$; it solves a nearby problem. The size of this nearby perturbation depends on the working precision, the problem dimensions, and the stability of the chosen factorization. That is why the choice between normal equations, QR, SVD, and later randomized surrogates is not only a question of flop counts but also of how perturbations are amplified through the conditioning of the underlying operator [2].

## 1.5 Symmetric Positive Definite Operators

The covariance-transport part of the thesis is built from symmetric positive semidefinite and positive definite matrices. A matrix $A \in \mathbb{R}^{p\times p}$ is symmetric positive semidefinite if

$$x^\top Ax \geq 0 \qquad \forall x \in \mathbb{R}^p,$$

and symmetric positive definite if the inequality is strict for every nonzero $x$. In the latter case, $A$ is invertible and all of its eigenvalues are strictly positive. Since the matrices are symmetric, they admit an orthogonal spectral decomposition

$$A = Q\Lambda Q^\top,$$

where $Q$ is orthogonal and $\Lambda$ is diagonal with real entries.

Empirical covariance matrices provide the main examples later on. If $Z \in \mathbb{R}^{n\times p}$ is a centered data matrix, then

$$C = \frac{1}{n} Z^\top Z$$

is symmetric positive semidefinite. In high-dimensional settings it may be singular or nearly singular, which is why regularization by $\lambda I$ is not an optional detail but a structural part of the method.

## 1.6 Matrix Square Roots and Regularization

If $A$ is symmetric positive definite with spectral decomposition $A = Q\Lambda Q^\top$, its matrix square root and inverse square root are defined by functional calculus as

$$A^{1/2} = Q\Lambda^{1/2}Q^\top, \qquad A^{-1/2} = Q\Lambda^{-1/2}Q^\top.$$

These matrices are again symmetric positive definite, and they are the unique positive definite solutions of

$$A^{1/2}A^{1/2} = A, \qquad A^{-1/2}AA^{-1/2} = I.$$

Regularization shifts the spectrum away from zero. If $C$ is positive semidefinite and $\lambda > 0$, then

$$C + \lambda I$$

is positive definite, with eigenvalues $\mu_i(C) + \lambda$. This shift provides a numerical safety margin for the inverse square root and controls how strongly small covariance eigenvalues are amplified.

## 1.7 Basic Perturbation Facts

The later covariance-transport analysis repeatedly uses simple spectral perturbation facts. The first is positivity under perturbation: if $A$ is symmetric positive definite and $E$ is symmetric, then

$$\lambda_{\min}(A + E) \geq \lambda_{\min}(A) - \|E\|_2.$$

Consequently, if $\|E\|_2 < \lambda_{\min}(A)$, the perturbed matrix remains positive definite. This is the basic condition that allows the square root and inverse square root of a computed approximation to remain well defined.

The second fact is qualitative rather than exact: matrix square roots and inverse square roots inherit the conditioning of the underlying spectrum. Perturbations of $A^{1/2}$ are controlled by the distance of the spectrum from zero, while perturbations of $A^{-1/2}$ are more sensitive because the scalar map $t \mapsto t^{-1/2}$ has derivative $-\frac{1}{2}\,t^{-3/2}$. This is the analytic source of the regularization tradeoff in covariance transport. Increasing $\lambda$ improves stability of the inverse square root, but it also changes the operator that is being transported.

## 1.8 Subspace Embeddings and Dimensionality Reduction

The idea of replacing a large least-squares problem by a smaller one rests on a geometric requirement: the sketching matrix should approximately preserve norms and angles on the relevant subspace. This requirement is captured by the notion of an oblivious subspace embedding (OSE).

**Definition 1.1** (Oblivious Subspace Embedding). Let $\epsilon \in (0, 1)$ and $\delta \in (0, 1)$. A distribution $\mathcal{S}$ over $\mathbb{R}^{s \times n}$ is a $(d, \epsilon, \delta)$-OSE if, for every fixed subspace $\mathcal{V} \subseteq \mathbb{R}^n$ of dimension at most $d$,

$$\Pr_{S \sim \mathcal{S}}\Big[\forall\, v \in \mathcal{V}:\ (1-\epsilon)\|v\|_2^2 \leq \|Sv\|_2^2 \leq (1+\epsilon)\|v\|_2^2\Big] \geq 1 - \delta.$$

The squared-norm convention is natural because it is equivalent to controlling the spectral norm of the Gram distortion $U^\top S^\top SU - I$ on an orthonormal basis $U$ of the embedded subspace. It immediately implies $\sqrt{1-\epsilon}\,\|v\|_2 \leq \|Sv\|_2 \leq \sqrt{1+\epsilon}\,\|v\|_2$.

The OSE property is a uniform Johnson–Lindenstrauss statement on subspaces. The Johnson–Lindenstrauss lemma itself provides the foundational concentration result for finite point sets:

**Lemma 1.2** (Johnson–Lindenstrauss). Let $\epsilon \in (0,1)$ and let $\mathcal{P}$ be a finite set of $p$ points in $\mathbb{R}^n$. If $S \in \mathbb{R}^{s\times n}$ has entries drawn independently from $\mathcal{N}(0,1/s)$ and

$$s \geq \frac{C\log p}{\epsilon^2},$$

then with probability at least $1-1/p$,

$$(1-\epsilon)\|u-v\|_2^2 \leq \|S(u-v)\|_2^2 \leq (1+\epsilon)\|u-v\|_2^2$$

for every pair $u,v \in \mathcal{P}$.

The passage from pointwise concentration (JL) to uniform subspace control (OSE) proceeds by a net argument: a $\gamma$-net of the unit ball in a $d$-dimensional subspace has cardinality at most $(3/\gamma)^d$; applying JL to this net and using the Lipschitz continuity of the sketching operator yields the subspace embedding property.

These definitions are used extensively in ?? → p. ??, where specific sketch families (Gaussian, Rademacher, SRHT) are shown to satisfy the OSE condition with concrete sketch-size bounds. The present section merely fixes the geometric language; the algorithmic and probabilistic details belong to the specialized chapters.

## 1.9 Why This Background Matters

These spectral facts extend the least-squares viewpoint from compressed rectangular operators to self-adjoint covariance operators. In the least-squares chapters, the decisive question is whether sketching preserves the geometry of a range space well enough that the reduced problem remains faithful to the original one. In the covariance-transport chapter, the decisive question becomes whether a randomized and finite-precision approximation of the covariance operator remains accurate enough after it is processed by matrix square roots and inverse square roots. The mathematical language is therefore the same in spirit, but the operator calculus is richer.

## 1.10 Least-Squares Perturbation Bounds

For the least-squares chapters, a more detailed perturbation statement is needed than the qualitative bound given earlier. The classical results of Wedin and Stewart provide the right framework.

**Theorem 1.3** (Wedin). Let $A, \tilde{A} \in \mathbb{R}^{n\times d}$ with $n \geq d$, and suppose $A$ has full column rank. Let $x_\star$ and $\tilde{x}_\star$ be the least-squares solutions of

$$\min_x \|Ax-b\|_2, \qquad \min_x \|\tilde{A}x-\tilde{b}\|_2.$$

Write $r_\star = b - Ax_\star$ and $\tilde{r}_\star = \tilde{b} - \tilde{A}\tilde{x}_\star$ for the optimal residuals. Then

$$\frac{\|\tilde{x}_\star - x_\star\|_2}{\|x_\star\|_2} \le \kappa_2(A)\left(\frac{\|\Delta A\|_2}{\|A\|_2} + \frac{\|\Delta b\|_2}{\|b\|_2}\right) + \kappa_2(A)^2\,\frac{\|r_\star\|_2}{\|A\|_2\,\|x_\star\|_2}\,\frac{\|\Delta A\|_2}{\|A\|_2} + O\big(\|\Delta\|^2\big),$$

where $\Delta A = \tilde{A} - A$, $\Delta b = \tilde{b} - b$, and $\|\Delta\|$ denotes the maximum relative perturbation magnitude.

The qualitative content is the same as the earlier schematic bound, but Theorem 1.3 →p.6 makes the constant structure explicit. The first term reflects the direct sensitivity of the solution to data perturbations, amplified by $\kappa_2(A)$. The second term reflects the additional sensitivity that arises when the problem is inconsistent: the residual $r_\star$ is not zero, and the perturbation of the residual direction is amplified by $\kappa_2(A)^2$.

This distinction between the consistent and inconsistent cases is central to the thesis. In sketch-and-solve methods, the sketch replaces the original operator by a compressed one. Sketching is not an additive perturbation of $A$ in the usual matrix sense, since $SA$ lives in a different row space and $SA - A$ is not a valid perturbation of compatible dimensions. Its effect is instead measured by how much the sketch distorts the geometry of the relevant subspace, equivalently by the Gram distortion $U^\top S^\top SU - I$, where $U$ is an orthonormal basis for the column space of $A$ (or the augmented space of $[A\ r_\star]$). This distortion is governed by the embedding parameter $\epsilon$, and its effect on the solution is filtered through $\kappa_2(A)$ and the residual size. Therefore, a small embedding distortion does not guarantee a small solution error unless the original problem is well-conditioned or nearly consistent. The experiments track both residual ratio and relative solution error precisely because these two quantities probe different aspects of the perturbation structure.

In the covariance-transport chapters, the same pattern reappears in a different algebraic form. The perturbation of the transport map $T_\lambda = A^{-1/2}B^{1/2}$ is not governed by $\kappa_2(A)$ alone, but by the conditioning of the regularized matrix functions $A^{-1/2}$ and $B^{1/2}$. The inverse-square-root map has a derivative proportional to $\lambda_{\min}^{-3/2}$, so the effective amplification factor is

$$K_\lambda(C_X, C_Y) \sim \frac{1}{\lambda_{\min}(C_X + \lambda I)^{3/2}}\,,$$

which is the direct analogue of $\kappa_2(A)^2$ in the least-squares setting. Regularization $\lambda$ plays the same structural role as improving the conditioning of the least-squares problem: it controls the amplification factor at the cost of changing the operator being approximated.

# References

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